

TOPLITA, Romania

WMO: 151070

Lat: 46.9264N

Lon: 25.3600E

Elev: 688

StdP: 93.33

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.8	(c) -17.0	(d) -20.5	(e) 0.7	(f) -18.6	(g) -18.2	(h) 0.8	(i) -15.8	(j) 9.2	(k) -6.5	(l) 7.4	(m) -5.2	(n) 0.4	(o) 130	(p) 0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	29.7	18.8	28.0	18.1	26.3	17.6	19.9	27.1	19.1	26.0	18.4	24.9	3.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.7	22.1	16.7	13.0	21.5	16.0	12.4	20.5	60.0	27.2	57.2	25.9	54.7	24.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.3	(b) 6.2	(c) 5.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-5.4	-2.9	1.9	7.3	12.6	16.5	17.6	17.9	12.5	7.4	1.7	-3.5
	DBStd	9.15	5.12	5.01	4.25	4.35	3.15	2.99	2.82	3.03	3.79	3.82	4.67	5.23
	HDD10.0	1991	478	362	251	103	12	1	0	0	18	98	251	417
	HDD18.3	4204	737	595	509	332	177	71	47	44	176	340	501	676
	CDD10.0	903	0	0	0	21	93	196	237	244	94	16	1	0
	CDD18.3	75	0	0	0	0	1	16	26	31	2	0	0	0
	CDH23.3	1237	0	0	0	9	41	248	357	499	81	1	0	0
	CDH26.7	314	0	0	0	0	4	56	93	148	13	0	0	0
Wind	WSAvg	1.7	1.3	1.4	1.8	2.2	2.0	2.0	2.0	1.7	1.6	1.5	1.2	1.3
Precipitation	PrecAvg	701	29	32	39	57	79	102	96	75	61	54	42	43
	PrecMax	971	53	69	94	93	145	182	173	121	148	106	101	121
	PrecMin	480	9	4	7	22	30	34	33	32	12	1	0	13
	PrecStd	117	13	16	20	23	29	36	39	25	32	28	24	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	10.2	18.6	24.5	27.0	30.5	31.3	32.5	28.2	22.3	15.1	8.3
		MCWB	3.4	6.5	10.2	13.1	15.5	19.6	19.4	19.1	16.9	13.8	10.7	6.2
	2%	DB	3.4	6.8	14.6	22.1	24.3	28.2	29.1	30.0	25.4	20.1	12.0	5.4
		MCWB	2.1	4.5	8.3	12.5	15.3	18.6	18.8	18.4	16.1	12.8	9.3	4.5
	5%	DB	1.8	4.7	11.3	19.6	22.3	26.2	27.2	28.2	23.1	17.3	9.8	3.6
		MCWB	0.9	2.8	6.4	11.5	14.7	17.9	18.4	18.2	15.5	11.7	7.6	2.8
	10%	DB	0.9	3.0	8.4	16.3	20.1	23.9	25.1	26.2	20.1	14.2	7.7	2.2
		MCWB	0.3	1.5	4.8	10.0	13.4	16.7	17.7	17.7	14.3	10.6	6.3	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	7.3	10.9	14.0	17.2	20.8	20.7	21.0	18.2	15.0	11.1	6.6
		MCDB	5.0	10.1	17.7	22.6	23.9	28.6	27.8	29.1	25.6	20.2	14.2	7.5
	2%	WB	2.2	5.0	8.7	13.0	16.1	19.5	19.7	19.8	17.0	13.6	9.6	4.3
		MCDB	3.0	6.6	14.1	21.0	22.6	26.5	26.9	26.7	23.9	18.5	12.0	5.1
	5%	WB	1.2	3.2	6.9	11.9	15.2	18.5	19.0	19.0	16.0	12.4	8.2	3.0
		MCDB	1.7	4.5	11.1	19.0	21.4	25.0	25.5	26.1	22.2	16.4	9.6	3.7
	10%	WB	0.3	1.8	5.2	10.6	14.1	17.5	18.2	18.2	14.9	10.9	6.6	1.6
		MCDB	0.9	2.6	7.8	15.7	19.1	23.3	24.1	25.1	19.7	13.8	7.8	2.1
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.1	11.0	12.9	12.9	13.3	13.4	14.9	12.6	12.4	7.9	6.3
		MCDBR	6.5	9.6	17.0	19.9	16.8	17.4	17.6	18.9	17.9	18.5	11.2	6.8
		MCWBR	5.0	7.1	10.5	10.4	8.4	7.5	7.8	8.0	8.8	11.1	7.8	5.6
	5% WB	MCDBR	5.7	9.1	16.5	17.9	15.5	15.8	15.2	16.3	16.4	16.7	10.0	6.9
	MCWBR	4.6	6.8	10.2	9.4	8.0	7.3	7.1	7.4	8.2	10.1	7.0	5.6	
Clear-Sky Solar Irradiance	taub	0.302	0.334	0.382	0.443	0.419	0.413	0.437	0.444	0.399	0.360	0.332	0.305	
	taud	2.303	2.225	2.174	2.064	2.211	2.265	2.201	2.177	2.268	2.366	2.378	2.348	
	Ebn at Noon	776	819	829	809	848	853	825	800	805	785	735	728	
	Edh at Noon	76	102	125	154	138	131	138	136	113	88	72	66	
All-Sky Solar Radiation	RadAvg	1.25	2.09	3.19	4.27	5.26	5.66	5.61	5.17	3.69	2.52	1.42	0.96	
	RadStd	0.17	0.23	0.39	0.42	0.50	0.47	0.50	0.47	0.44	0.32	0.19	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		+0.70	N/A	N/A	+0.85	N/A	N/A	-158	-220	+83	+19

Nomenclature: See separate page